

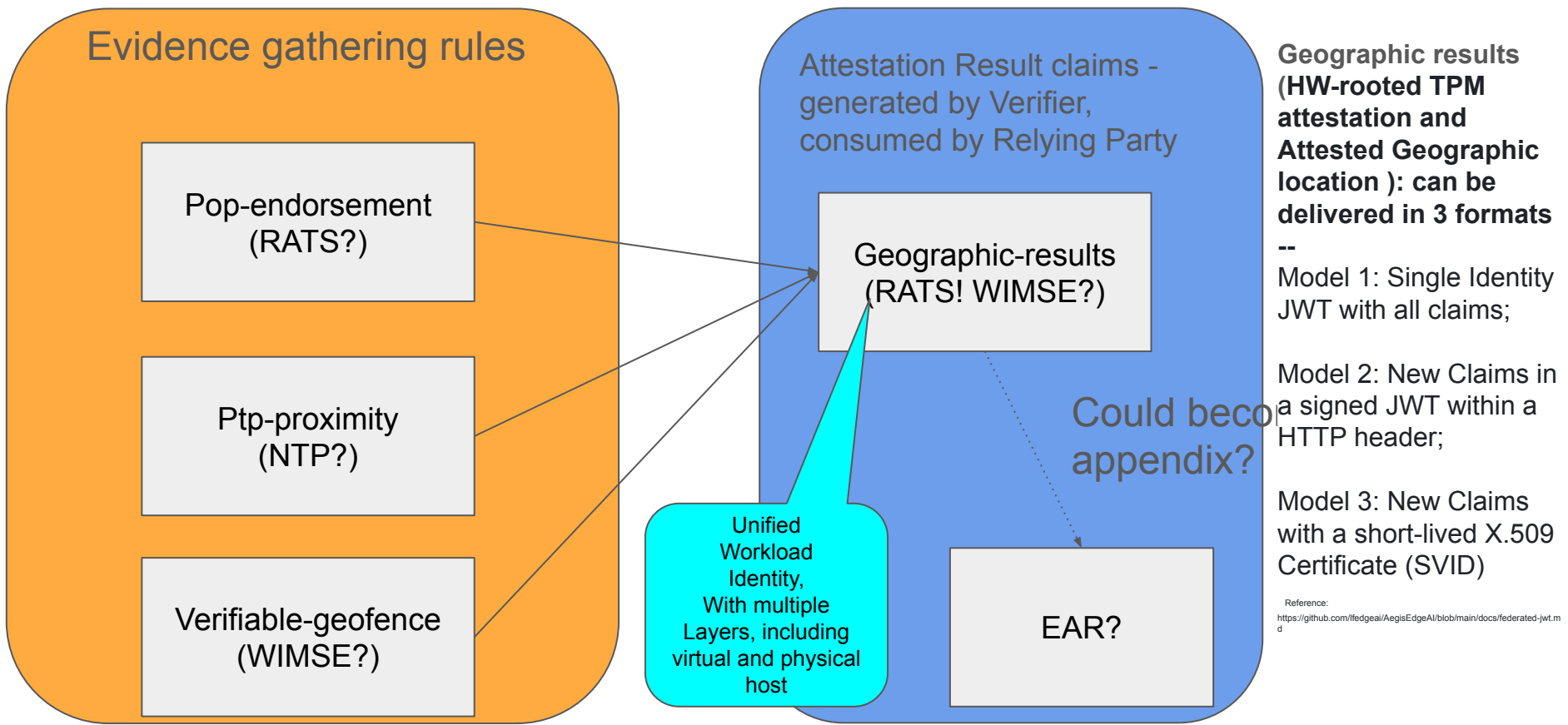
Remote Attestation And Geographic Evidence/Results

IETF 124 - Montreal
RATS and WIMSE WG
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Relevant Document(s)

- <https://datatracker.ietf.org/doc/draft-richardson-rats-geographic-results/> (new)
- [draft-lkspa-wimse-verifiable-geo-fence](#) (discussed at IETF123: RATS+WIMSE)
- Using PTP to estimate proximity - <https://github.com/ramkri123/ptp-asymmetric-authentication> (new)
- <https://datatracker.ietf.org/doc/draft-richardson-rats-pop-endorsement/> (discussed at IETF123: RATS)

Possible Document Architecture/Combinations: many IETF docs



Document Architecture/Combinations: fewer Documents, many appendix

Attestation Result claims - generated by Verifier, consumed by
Relying Party

geographic-results

Appendix: Evidence gathering rules

verifiable-geofence

ptp-proximity

pop-endorsement

**Geographic results
(HW-rooted TPM
attestation and
Attested Geographic
location): can be
delivered in 3 formats**

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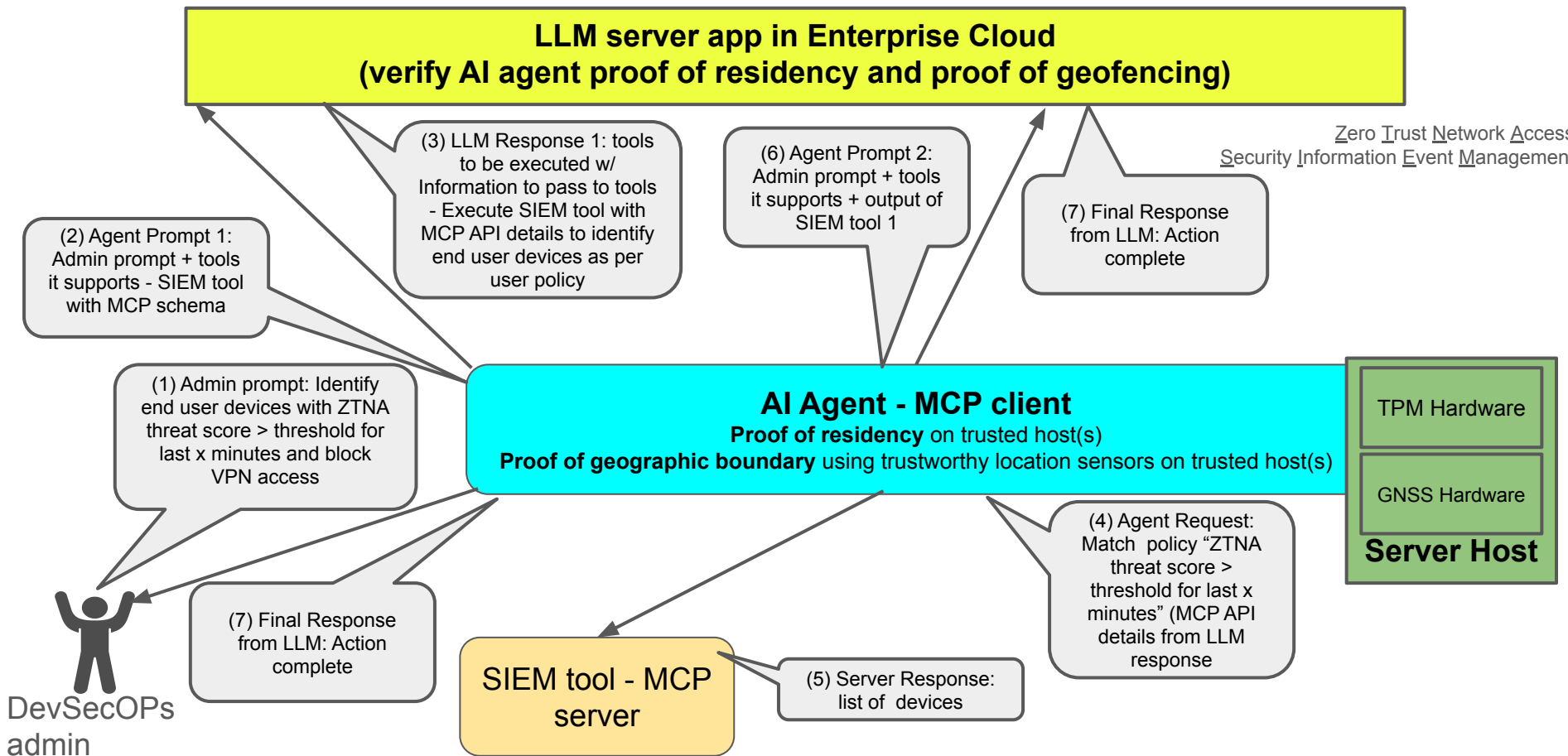
Model 1: Single Identity
JWT with all claims;

Model 2: New Claims in
a signed JWT within a
HTTP header;

Model 3: New Claims
with a short-lived X.509
Certificate (SVID)

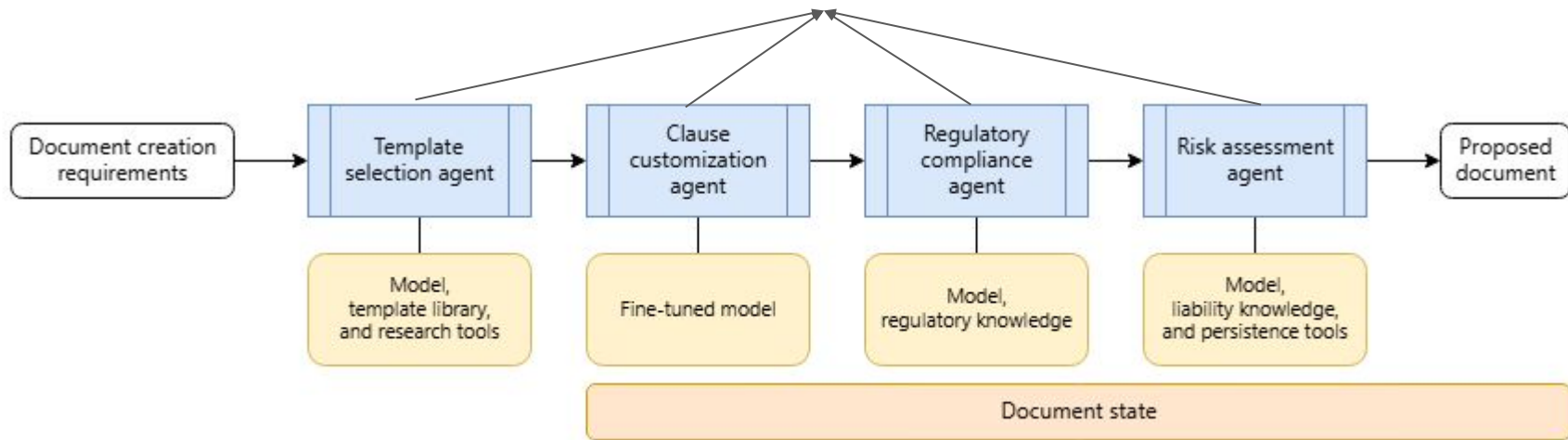
Reference:
<https://github.com/lfedgeai/AegisEdgeAI/blob/main/docs/federated-jwt.md>

Use Case: An AI Agent (Devsecops admin persona) is assessing the trustworthiness of enterprise VPN users.
The Agent's geography is important, because it will be provided with sensitive security information



Multi-agent Use Case - Verifiable host residency & geofencing

Trusted Agent Registry



Reference: <https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns>

Example contract use case: Ensure the contract processing matches the geolocation policy -- for instance, different consumer protection laws (GDPR, HIPPA, Local laws etc.) are geographic region specific.

Solution: Each agent in the Enterprise supply chain needs to provide to the trusted agent registry

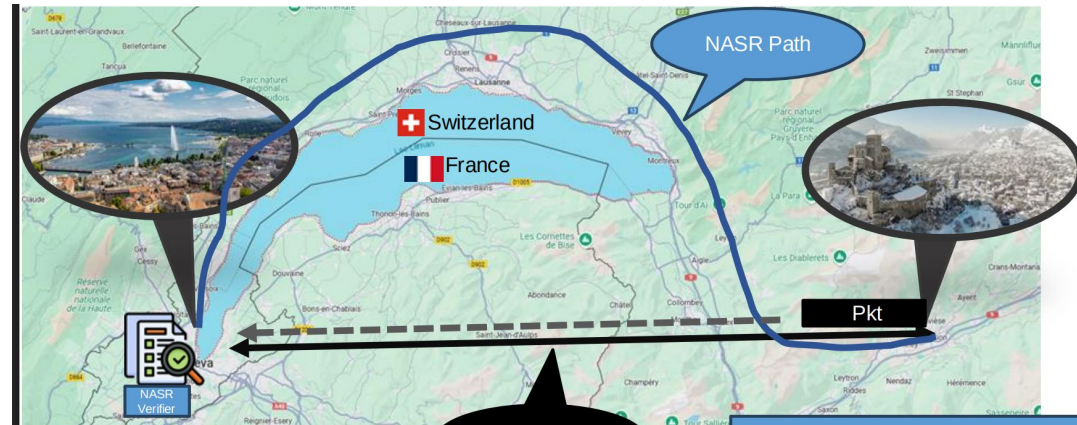
- Cryptographically verifiable proof of residency on trusted hosts
- Cryptographically verifiable proof of geofencing on trusted hosts with trusted sensors

Use Case: Network Attested Secure foRwarding (NASR)

Requires Evidence or
Endorsement as to
residency of forwarding
elements:

- Might be sovereignty
- Might be performance/latency
- Might be redundancy/resilience

Below is an example where
The shortest path would be via a different country.



Considerations

- Different groups will have different needs, and putting too much together may confuse and/or slow review process
- But, a goal is to simplify the effort by Relying Parties, to have common code
 - Thus a common Attestation Results, arrived at by many different procedures

Discussion

Before looking
for adoption,
We need to
decide on
structure.

